Market Survey

FLEXIBLE, PRINTED AND ORGANIC ELECTRONICS: The Bus India Should Not Miss!

Sudeshna Das is director at ComConnect Consulting

Continuous innovation and development of advanced organic electronic technologies is likely to augment the growth of the global organic electronics market in the near future

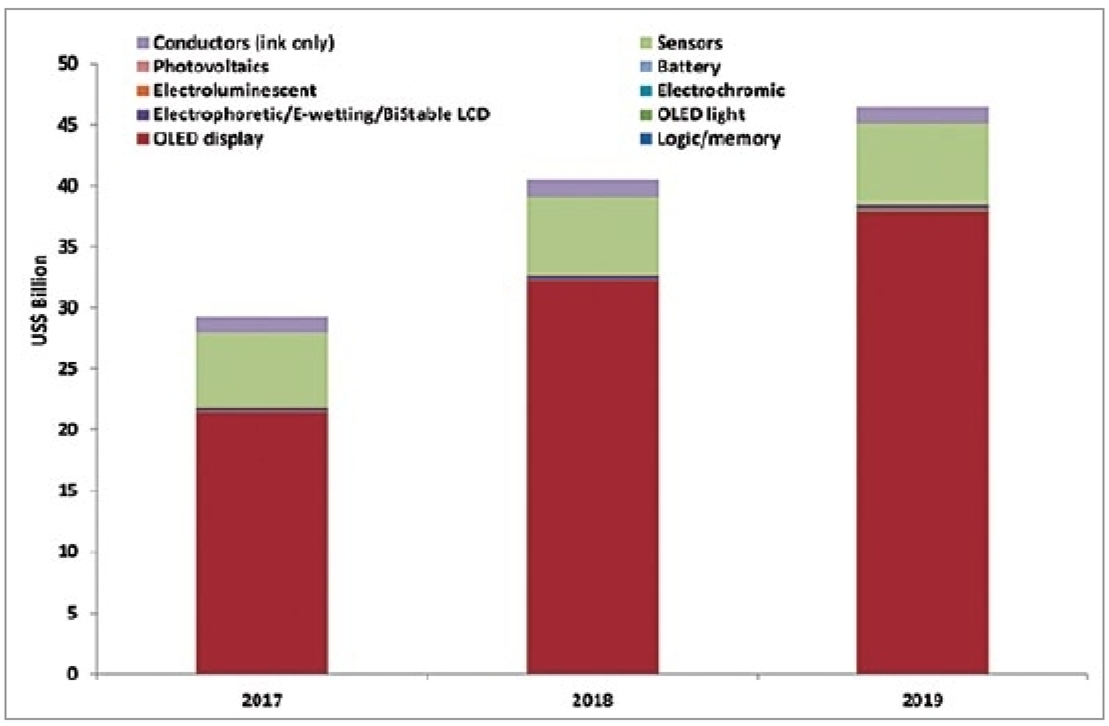
The need for innovative products made from non-toxic materials at an affordable price brought about disruptive changes in the electronics sector. It also catalysed the evolution of microelectronics to flexible, printed and organic electronics. Emergence of this new segment is being driven by the use of new materials and methods of manufacturing. This break from the past has provided an opportunity to countries like India who missed the microelectronics bus earlier.

Market potential

According to Markets And Markets, the flexible electronics market was valued at US$ 17.85 billion in 2017. It is expected to be worth US$ 40.37 billion by 2023, at a CAGR of 11 per cent between 2018 and 2023.

IDTechEx has a much more optimistic forecast. It states that the total market for printed, flexible and organic electronics will grow from US$ 29.28 billion in 2017 to US$ 73.43 billion in 2027. This is shown in the figure below.

Majority of the flexible electronics market is OLEDs (organic but not printed) and conductive ink used for a wide range of applications. On the other hand, products such as stretchable electronics, logics and memories, and thin-film sensors constitute a much smaller segment, but with huge growth potential.



Market forecast by component type in US$ billions

Flexible electronics allow the development of new applications by integrating intelligence in the form of electronics directly on flexible substrates such as plastics, paper, textiles or metal foils. This opens up new possibilities of developing conformal, flexible, lighter and more robust applications. The core of these applications is electronics that can be printed roll-to-roll using fast manufacturing processes, much in the same way newspapers are printed, thereby, making the products much more affordable and, if required, disposable.

Large-area flexible electronics are on the threshold of an imminent revolution that is driven by innovative applications made possible at a much lower cost than the conventional methods of manufacturing. At the heart of this revolution lie two significant capabilities: designing products that are flexible and form fitting, and their manufacturing by printing-based processes. The combination of the two may lead to roll-to-roll, large volume and high throughput manufacturing.

Organic electronics is useful for high-performance devices in four major application areas: displays, photovoltaic, lighting and integrated smart systems. These are briskly growing into a scientific and technological revolution, and are expected to have a large impact on our lives as compared to microelectronics.

Applications at a glance

Applications of flexible electronics are wide ranging and span several sectors. These include smart packages and labels for brand protection and anti-counterfeiting, wearables and lightweight electronics, distributed energy production through organic solar cells on curved surfaces, dis-